Lecture 5: Atmospheres I
Composition, Structure, & Energy Balance

Planetary Systems

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Earth’s thin blue line, ISS STS-129 (2009). Credit: NASA

Learning objectives

By the end of this lecture, you will be able to:

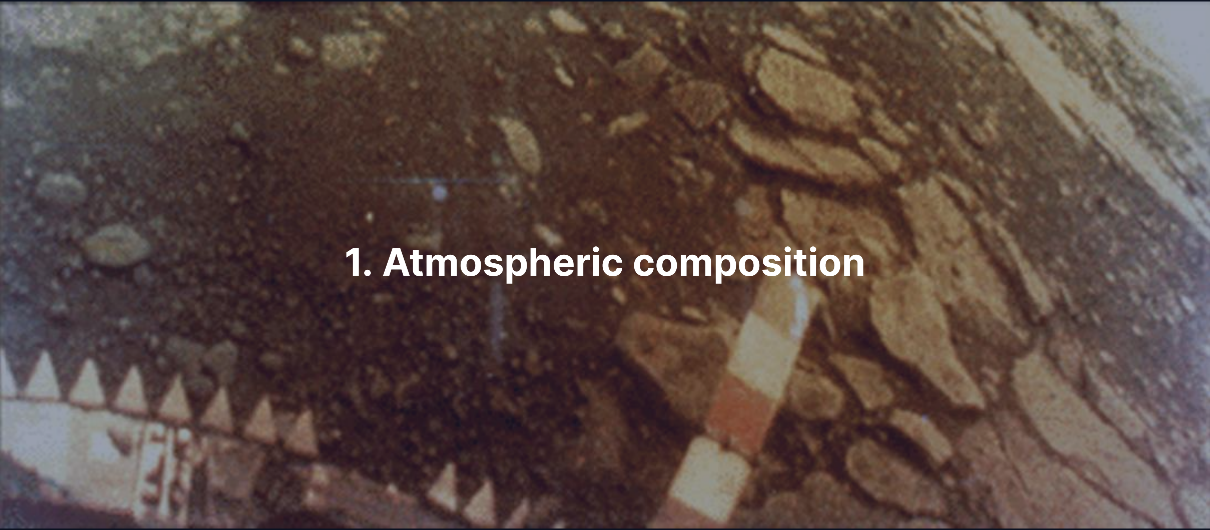
- Classify atmospheric types (primary, secondary, tertiary)
- Derive pressure profiles from hydrostatic equilibrium
- Compute the atmospheric scale height and apply it across the solar system
- Explain how the greenhouse effect raises surface temperature above T_{eff}
- Evaluate atmospheric escape mechanisms and retention criteria

Recap: Lecture 4

- Metal-silicate separation in magma oceans drove core formation within ~30–60 Myr
- Dynamo requires conducting fluid, convection, and magnetic Reynolds number $\mathrm{Rm} \gtrsim 10\text{–}100$
- Mars lost its dynamo between ~4.1 and 3.7 Ga, triggering atmospheric erosion

Today: The thin gaseous envelope that controls a planet's surface conditions.

1. Atmospheric composition



The surface of Venus under its 92-bar atmosphere, Venera 13 (1982). Credit: USSR/NASA NSSDC, public domain

Primary atmospheres

Captured directly from the protoplanetary disk.

- Dominated by $\mathrm{H_2}$ and He (solar composition)
- Requires core mass ≳ 5–10 $M_{\oplus}$ before disk dispersal in ~3–10 Myr
- **Gas giants:** Jupiter and Saturn retain massive $\mathrm{H_2}$/He envelopes
- **Ice giants:** Uranus and Neptune envelopes are only 10–20% of total mass
- Terrestrial planets are too small to retain primordial $\mathrm{H_2}$/He

Secondary atmospheres

Produced by outgassing from the interior.

- Magma ocean degassing and volcanism release dissolved volatiles
- Speciation depends on mantle **oxygen fugacity** (f_{O_2} redox state):
  - Oxidising: CO_2, H_2O, N_2
  - Reducing: H_2, CO, N_2
- **Venus:** CO_2-dominated (96.5%), 92 bar
- **Mars:** CO_2-dominated (95%), 6 mbar
- **Titan:** thick N_2 atmosphere (from NH_3), 1.5 bar

Tertiary atmospheres

Earth's atmosphere is a tertiary atmosphere produced by billions of years of biological modification.

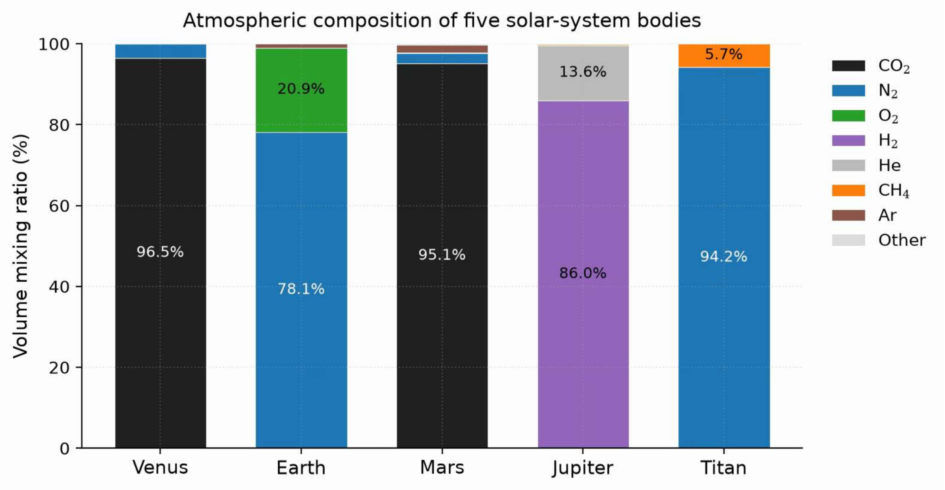
- Original outgassed inventory was dominated by CO_2 and N_2, similar to Venus
- Oxygenic photosynthesis added the O_2 (21%); the N_2 (78%) is outgassed and accumulated
- Atmospheric O_2 is entirely biogenic and requires active life to persist

Earth's O_2, and so its tertiary character, is continuously maintained by life.

Comparative atmospheric properties

	Venus	Earth	Mars	Jupiter	Titan
P_{surf} (bar)	92	1.0	0.006	n/a	1.5
T_{surf} (K)	737	288	215	n/a	94
Dominant gas	$\mathrm{CO_2}$	$\mathrm{N_2}$	$\mathrm{CO_2}$	$\mathrm{H_2}$	$\mathrm{N_2}$
Secondary gas	$\mathrm{N_2}$	$\mathrm{O_2}$	$\mathrm{N_2}$	He	$\mathrm{CH_4}$
μ	43.4	28.97	43.3	2.2	28.6
Type	2°	3°	2°	1°	2°

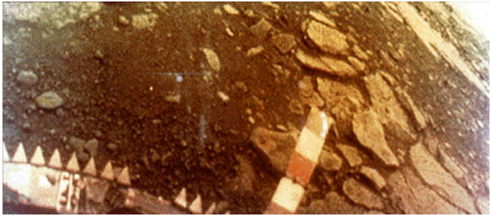
Staggering diversity: surface pressures spanning five orders of magnitude, temperatures from 94 to 737 K.



Atmospheric composition by volume for five solar-system bodies: three origins yield five distinct outcomes across the solar system. Course figure.

Venus: a secondary atmosphere in extremis

A massive secondary atmosphere transforms planetary surface conditions to extreme pressures and temperatures.



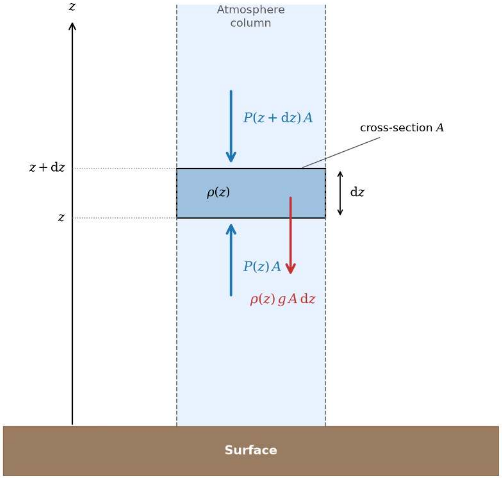
Credit: USSR/NASA NSSDC, public domain

- *Venera 13* landed 1 March 1982 and survived 127 minutes
- Surface pressure of 92 bar and surface temperature of 737 K
- Dense CO_2 atmosphere and cloud deck filter sunlight to a dim orange glow

2. Hydrostatic equilibrium

The thin atmosphere of Mars above its limb, Viking 1 (1976). Credit: NASA/JPL

Hydrostatic equilibrium



Credit: Course figure

Net vertical force on a thin slab of area A and thickness $\mathrm{d}z$ vanishes:

$$P(z)A - P(z + \mathrm{d}z)A = \rho\, g\, A\, \mathrm{d}z$$

$$\boxed{\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g}$$

Pressure decreases upward because each layer supports the weight of all gas above it.

The ideal gas law and barometric formula

The ideal gas law relates density to pressure: $P = \rho\, k_\mathrm{B} T / (\mu\, m_u)$.

Substituting $\rho(P)$ into hydrostatic equilibrium $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g$:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B} T} P = -\frac{P}{H}$$

For an **isothermal** atmosphere (constant T), integrating yields:

$$\boxed{P(z) = P_0 \exp\left(-\frac{z}{H}\right)}$$

where $H = k_\mathrm{B} T / (\mu\, m_u\, g)$; pressure drops by $e \approx 2.718$ each scale height.

3. Blackboard derivation: scale height

Sunrise over Earth’s limb from orbit. Credit: NASA

Deriving the scale height

Goal: Derive $H = k_\mathrm{B} T / (\mu\, m_u\, g)$ from hydrostatic equilibrium and the ideal gas law.

Start from:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho\, g \qquad \text{and} \qquad P = \frac{\rho\, k_\mathrm{B} T}{\mu\, m_u}$$

Express ρ in terms of P:

$$\rho = \frac{P\, \mu\, m_u}{k_\mathrm{B} T}$$

Substitute:

$$\frac{\mathrm{d}P}{\mathrm{d}z} = -\frac{\mu\, m_u\, g}{k_\mathrm{B} T} P \equiv -\frac{P}{H}$$

The atmospheric scale height

$$\boxed{H = \frac{k_{\mathrm{B}} T}{\mu\, m_u\, g}}$$

Physical interpretation:

- **Higher** T → larger H: hotter gas extends further
- **Heavier molecules** (larger μ) → smaller H: harder to loft
- **Stronger gravity** g → smaller H: more compressed

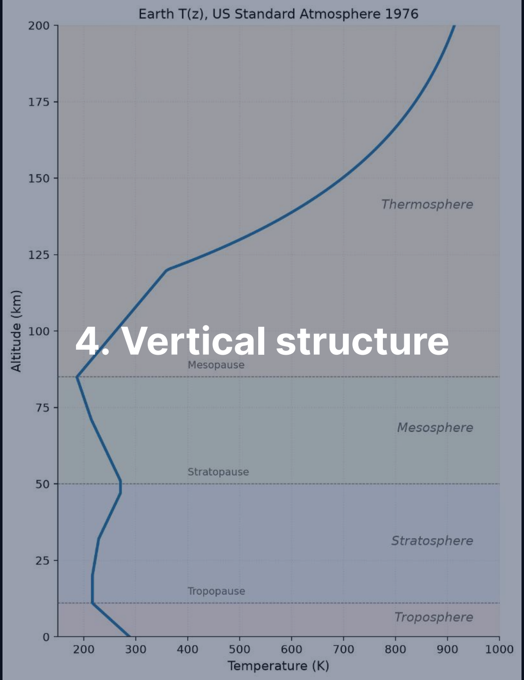
Key insight: H encodes the competition between thermal energy and gravitational binding; integrated over the whole column, the surface pressure is the column weight, $P_0 = m_{\mathrm{col}}\, g$.

Scale heights across the solar system

Body	T **(K)**	μ	g **(m s^{-2})**	H **(km)**
Venus	737	43.4	8.87	15.9
Earth	288	28.97	9.81	8.4
Mars	215	43.3	3.72	11.1
Jupiter	165	2.2	24.8	25
Titan	94	28.6	1.35	20

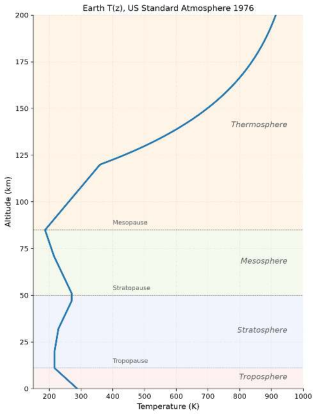
- Jupiter: large H despite strong gravity, because $\mathrm{H_2}$ is very light (μ = 2.2)
- Titan: large H despite low T, thanks to very weak gravity (g = 1.35 m s^{-2})
- Venus: hot but heavy molecules and decent gravity yield moderate H

4. Vertical structure



Earth’s atmospheric layers. Course figure

Atmospheric layers



Credit: Course figure; US Standard Atmosphere 1976

Layers defined by the sign of $\frac{\mathrm{d}T}{\mathrm{d}z}$:

1. **Troposphere** (0–12 km): T decreases with z; convective
2. **Stratosphere** (12–50 km): T increases; UV ozone heating
3. **Mesosphere** (50–85 km): T decreases to mesopause (~190 K)
4. **Thermosphere** (>85 km): T increases; EUV heating (T > 1000 K)

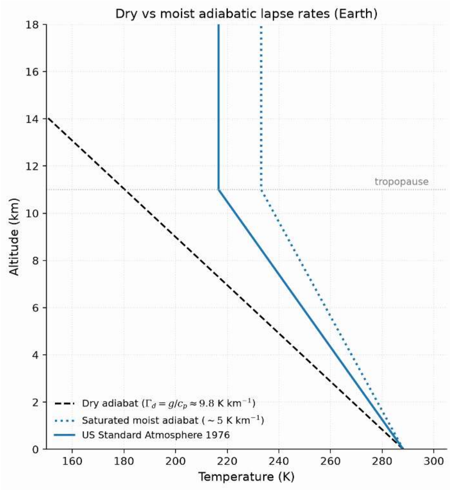
The adiabatic lapse rate

In the troposphere, convection sets the temperature gradient:

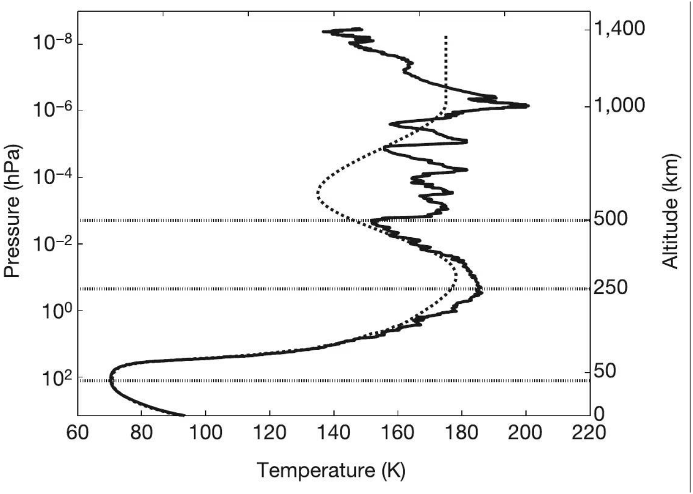
$$\Gamma_d = -\frac{\mathrm{d}T}{\mathrm{d}z} = \frac{g}{c_p}$$

- **Dry adiabatic lapse rate** for Earth: $\Gamma_d = 9.81/1004 \approx 9.8\ \mathrm{K\,km^{-1}}$
- Observed average: ~6.5 $\mathrm{K\,km^{-1}}$ (lower due to latent heat release)
- Physical picture: rising air expands, does work on surroundings, cools

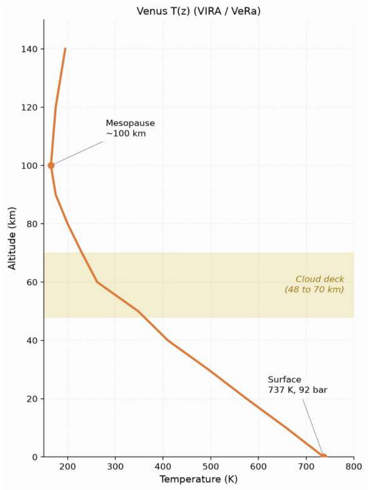
Tropopause height: ~8 km (poles) to ~17 km (equator).



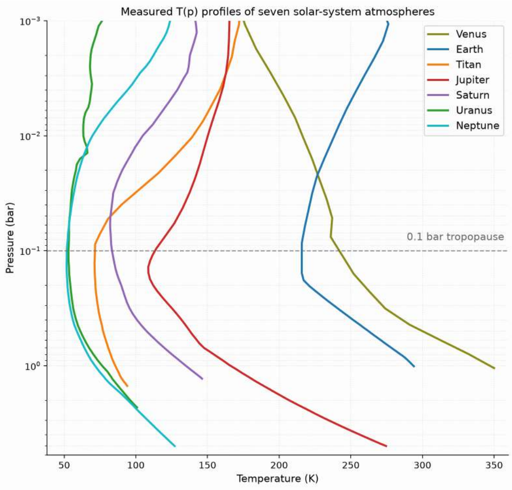
Dry and moist adiabats against the US Standard Atmosphere 1976 reference profile: latent heat makes the moist adiabat shallower, and real convection sits between both limits. Course figure.



Titan’s temperature profile from the Huygens HASI descent: an in situ descent measured the 94 K surface, 0.1 bar tropopause, and stratospheric structure. Fulchignoni et al. (2005).

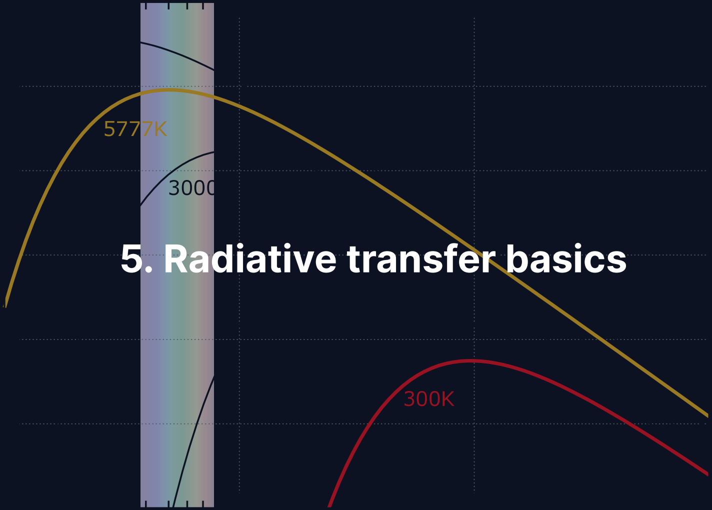


Venus temperature profile, Pioneer Venus/VIRA and Venus Express: without ozone, temperature falls monotonically from 737 K at the surface to the mesopause, reaching 1 bar near 50 km. Course figure.



Measured temperature-pressure profiles of seven solar-system atmospheres: thick atmospheres transition from convective to radiative near 0.1 bar, setting a universal tropopause minimum. Digitized from Robinson & Catling (2014), Fig. 1.

5. Radiative transfer basics



Blackbody spectra. Credit: Wikimedia, public domain

Three interactions: radiation and matter

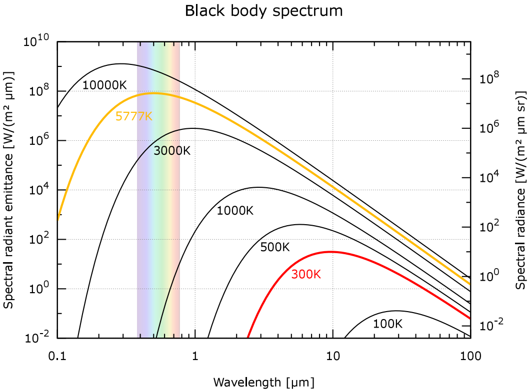
When radiation passes through an atmosphere:

Absorption: Photon energy converted to internal molecular energy
Key absorbers: $\mathrm{CO_2}$, $\mathrm{H_2O}$, $\mathrm{CH_4}$, $\mathrm{O_3}$, $\mathrm{N_2O}$

Emission: Warm gas radiates at absorbed wavelengths (Kirchhoff's law)
Radiates energy directly to space

Scattering: Photons redirected without absorption
Rayleigh scattering ($\propto \lambda^{-4}$) and Mie scattering

Stellar vs. planetary emission



Credit: Wikimedia, public domain

- Sun (~5800 K): peak at ~0.5 μm (visible)
- Planet (~300 K): peak at ~10 μm (thermal IR)

This **wavelength separation** is the physical basis of the greenhouse effect: the atmosphere can be transparent at visible wavelengths but opaque in the infrared.

Optical depth and the Beer-Lambert law

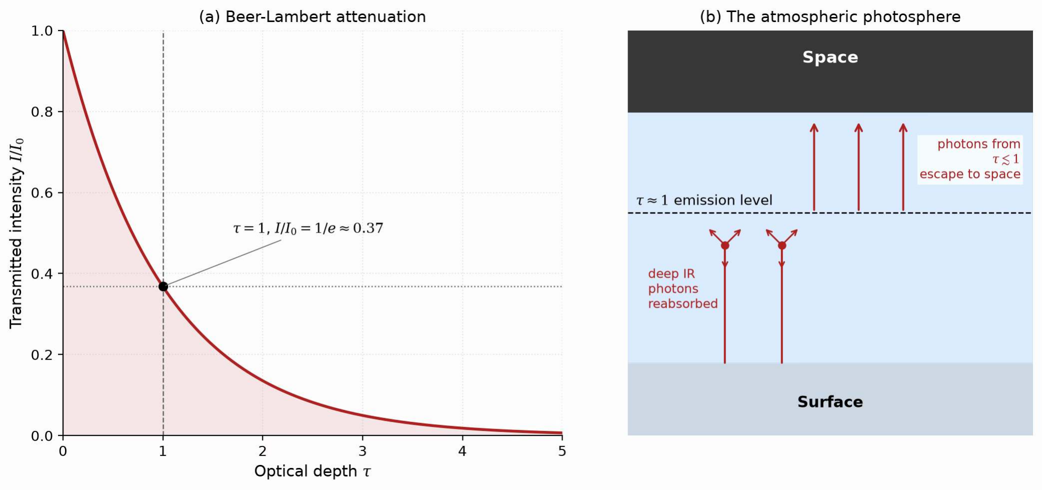
Optical depth τ measures cumulative absorption along a path:

$$\tau = \int_0^s \kappa \, \rho \, \mathrm{d}s'$$

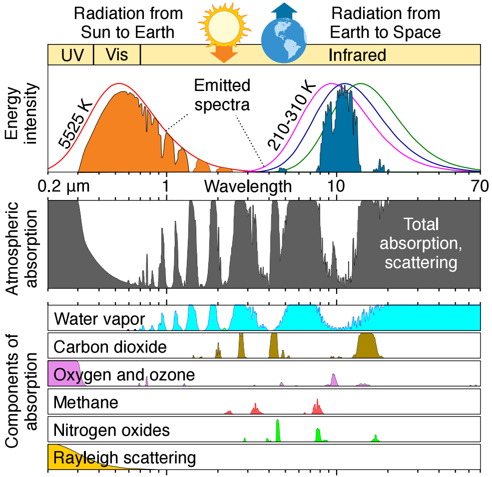
Beam intensity decays following the **Beer-Lambert law**:

$$\boxed{I = I_0 \, e^{-\tau}}$$

- **Optically thin** ($\tau \ll 1$): radiation passes freely through the gas
- **Optically thick** ($\tau \gg 1$): photons are repeatedly absorbed and re-emitted
- **Photosphere** ($\tau \approx 1$): thermal radiation escapes directly to space



The atmospheric photosphere: thermal radiation escapes to space from the $\tau \approx 1$ level, setting the effective temperature. Course figure.

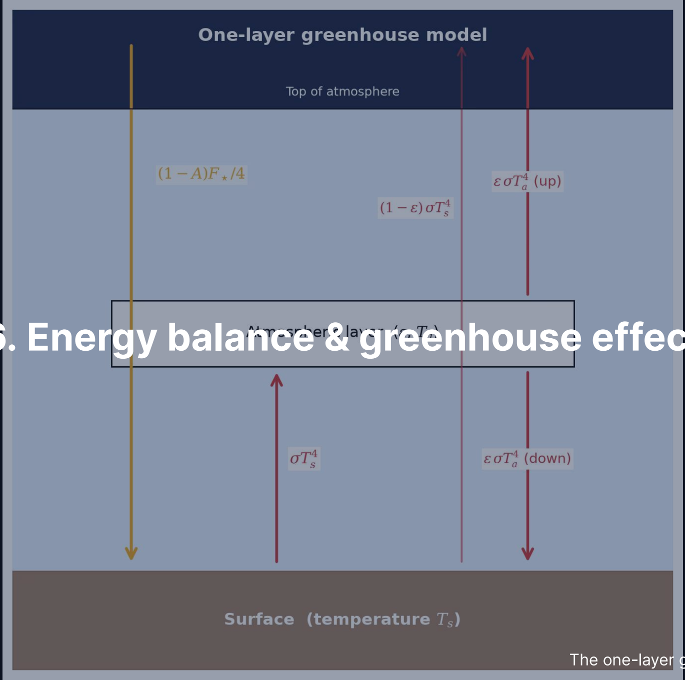


Earth's atmospheric absorption spectrum from UV to IR: greenhouse gases absorb outgoing surface infrared while leaving visible sunlight and the 8–13 μm window transparent. Credit: Wikimedia Commons, public domain.

Break

Earthrise, Apollo 8 (1968). Credit: NASA

6. Energy balance & greenhouse effect



The one-layer greenhouse model. Course figure

Radiative-convective equilibrium

Atmospheres divide into a convective lower layer and a radiatively controlled upper layer.

- **Troposphere**: greenhouse opacity drives convective instability along the adiabat
- **Stratosphere**: optically thin gas reaches radiative equilibrium, stable against convection
- **Tropopause**: boundary where the radiative temperature gradient becomes shallower than the adiabat

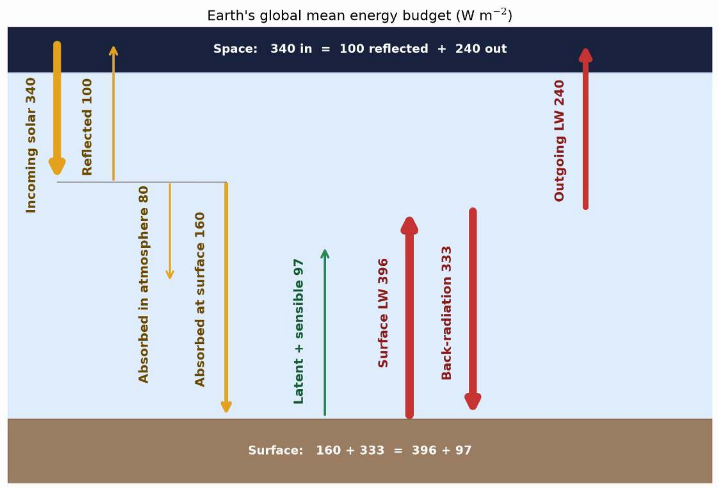
Planetary energy balance

Absorbed stellar power equals emitted thermal radiation:

$$(1 - A)\frac{S}{4} = \sigma T_{\mathrm{eff}}^4 \qquad \text{where} \quad S = \frac{L_\star}{4\pi d^2}$$

- A: Bond albedo; S: stellar flux ($S_0 = 1361\ \mathrm{W\,m^{-2}}$ at Earth)
- Factor $1/4$: ratio of intercepting cross-section πR_p^2 to emitting sphere $4\pi R_p^2$

$$\boxed{T_{\mathrm{eff}} = \left[\frac{(1 - A)\,L_\star}{16\pi\sigma\, d^2}\right]^{1/4} = \left[\frac{(1 - A)\,S}{4\sigma}\right]^{1/4}}$$



Earth's globally averaged energy budget in $\mathrm{W\,m^{-2}}$: atmospheric back-radiation of 333 $\mathrm{W\,m^{-2}}$ supplies more energy to the surface than direct sunlight (160 $\mathrm{W\,m^{-2}}$). After Trenberth et al. (2009).

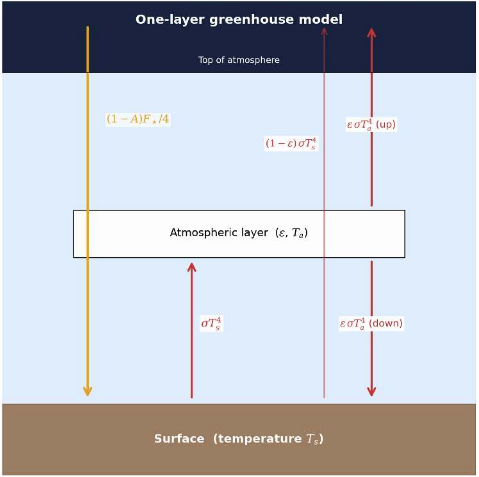
Effective vs. surface temperature

Body	A	d (AU)	T_{eff} (K)	T_{surf} (K)	ΔT (K)
Venus	0.77	0.72	227	737	+510
Earth	0.30	1.00	255	288	+33
Mars	0.25	1.52	210	215	+5
Jupiter	0.34	5.20	110	165*	+55

*Jupiter 1-bar level; excess partly from internal heat (Lecture 3).

$\Delta T = T_{\mathrm{surf}} - T_{\mathrm{eff}}$ reveals the **greenhouse effect**.
Venus: 510 K. Earth: 33 K. Mars: ~5 K (only 6 mbar, no water-vapour amplifier).

The greenhouse mechanism



Credit: Course figure

1. Sunlight (visible, ~0.5 μm) reaches the surface
2. Surface emits thermal IR (~10–15 μm)
3. Greenhouse gases absorb outgoing IR
4. Absorbing layer re-emits: half up, half down
5. Downward emission adds energy to surface

Result: $T_{\rm surf} > T_{\rm eff}$

One-layer greenhouse model

Single atmospheric layer with infrared emissivity ε, transparent to visible sunlight:

Flux balance:

Atmosphere: $\varepsilon\sigma T_s^4 = 2\varepsilon\sigma T_a^4$

Surface: $(1 - A)\frac{F_\star}{4} + \varepsilon\sigma T_a^4 = \sigma T_s^4$

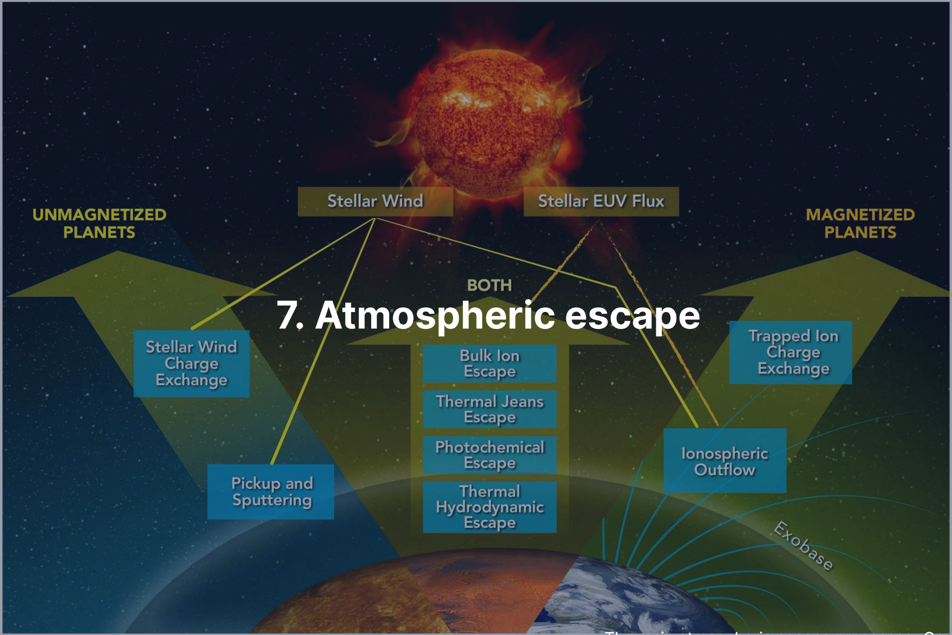
TOA: $(1 - A)\frac{F_\star}{4} = \sigma T_{\text{eff}}^4$

Surface temperature:

$$\boxed{T_s = T_{\text{eff}}\left(\frac{2}{2 - \varepsilon}\right)^{1/4}}$$

- $\varepsilon = 0$ (transparent): $T_s = T_{\text{eff}}$
- $\varepsilon = 1$ (opaque): $T_s = 2^{1/4}\, T_{\text{eff}} \approx 1.19\, T_{\text{eff}}$ (Earth: ≈303 K)

7. Atmospheric escape



The main atmospheric escape processes. Gronoff et al. (2020)

Thermal (Jeans) escape

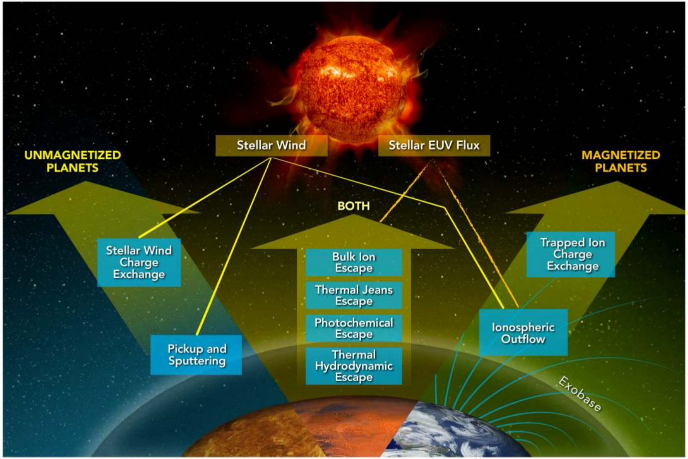
Molecules follow a Maxwell-Boltzmann distribution whose high-velocity tail can exceed escape speed.

The **Jeans escape parameter** at the exobase compares gravitational to thermal energy:

$$\lambda_J = \frac{G M_p \, m}{k_\mathrm{B} T_\mathrm{exo} \, r_\mathrm{exo}} = \frac{v_\mathrm{esc}^2}{v_\mathrm{th}^2}$$

with $v_\mathrm{th} = \sqrt{2 k_\mathrm{B} T_\mathrm{exo} / m}$ the most probable thermal speed.

- $\lambda_J \gg 1$: escape is negligible because the tail is depleted
- $\lambda_J \lesssim 2$–3: thermal escape drives rapid atmospheric loss
- Escape flux $\Phi_J \propto (1 + \lambda_J)\, e^{-\lambda_J}$ depends exponentially on λ_J



The main atmospheric escape processes: thermal, photochemical, and ion channels operate on all planets, while planetary magnetic fields select which pathways dominate. Gronoff et al. (2020).

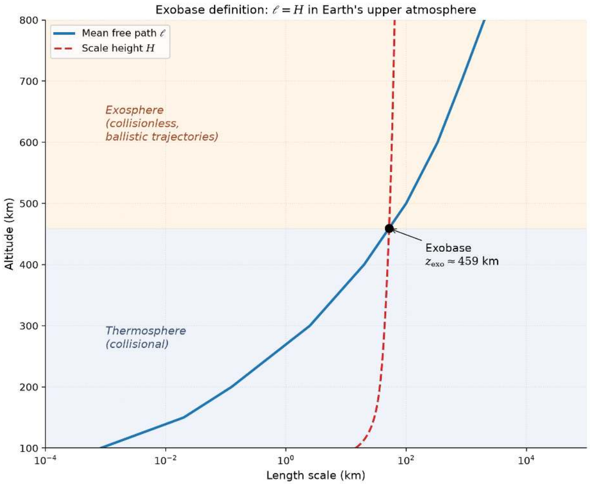
Jeans parameter for Earth and Mars

Because $\lambda_J \propto m$, light hydrogen escapes steadily while heavy molecules remain gravitationally bound.

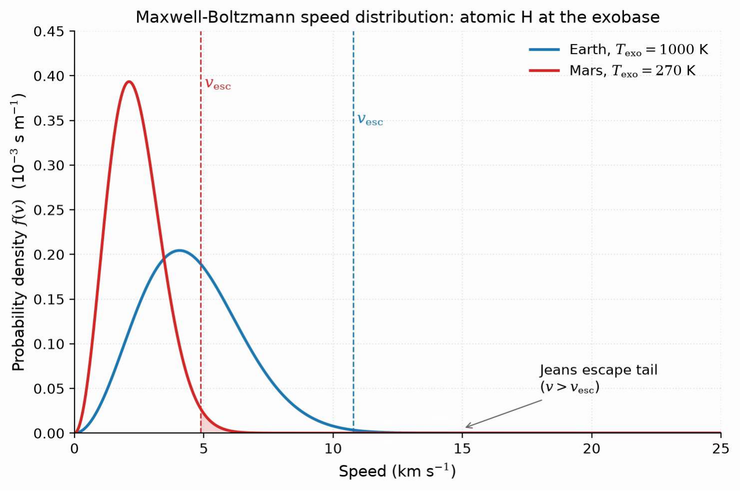
Exobase temperatures: 1000 K (Earth), 270 K (Mars).

Species	m **(u)**	λ_J **(Earth)**	λ_J **(Mars)**
H	1	7.0	5.3
H_2	2	14	11
He	4	28	21
N_2	28	200	150
CO_2	44	310	230

- $\lambda_J \gg 1$ for N_2 and CO_2: thermal escape is completely negligible
- Moderate λ_J allows atomic H to escape steadily from both planets



The exobase: altitude where the mean free path ℓ equals the scale height H, splitting collisional fluid below from ballistic flight above. Course figure.



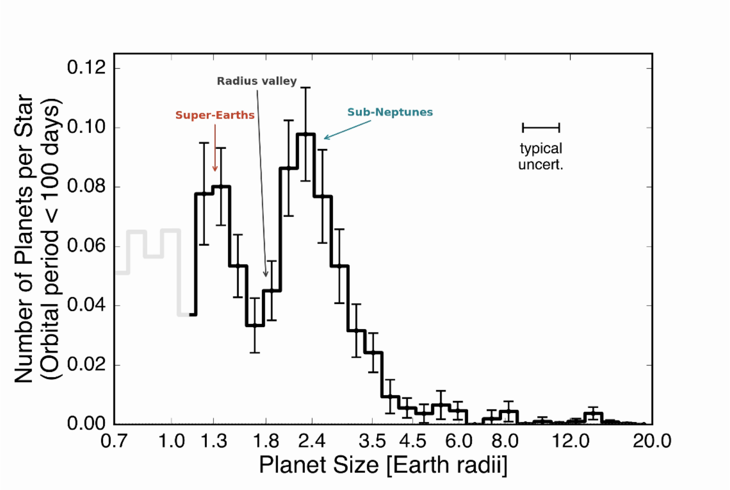
Maxwell-Boltzmann speed distributions for atomic H at Earth and Mars exobases: the escaping fraction sits in the high-velocity tail exceeding v_{esc}, scaling as $e^{-\lambda_J}$. Course figure.

Hydrodynamic escape

Intense stellar EUV heating drives bulk outflow as an **atmospheric wind**:

- Peak importance in first few hundred Myr when stellar EUV is 10–100× higher
- Strips H_2-rich primary envelopes from planets up to several $M_\oplus$
- Outflowing light hydrogen can drag along heavier volatiles (He, C, N, O)
- Primary mechanism sculpting the exoplanet radius valley at ~1.5–2 $R_\oplus$

Source: Hunten et al. (1987); Owen (2019)



The observed radius valley in the California-Kepler Survey: a bimodal distribution with a deficit near 1.8 $R_\oplus$ separates stripped rocky cores from envelope-retaining sub-Neptunes. After Fulton et al. (2017).

Energy-limited escape rate

Stellar EUV energy absorbed in the upper atmosphere drives hydrodynamic mass loss in the energy-limited regime.

$$\dot{M}_{\mathrm{EL}} = \eta \frac{\pi F_{\mathrm{EUV}} R_p^3}{G M_p}$$

- Efficiency $\eta \approx 0.1$–0.2 converts incident EUV flux into expansion work
- Scales as R_p^3: extended, low-density atmospheres lose mass most rapidly
- Scales inversely with M_p: lower planetary gravity enables stronger bulk outflow

Source: Watson et al. (1981); Owen (2019)

Non-thermal escape mechanisms

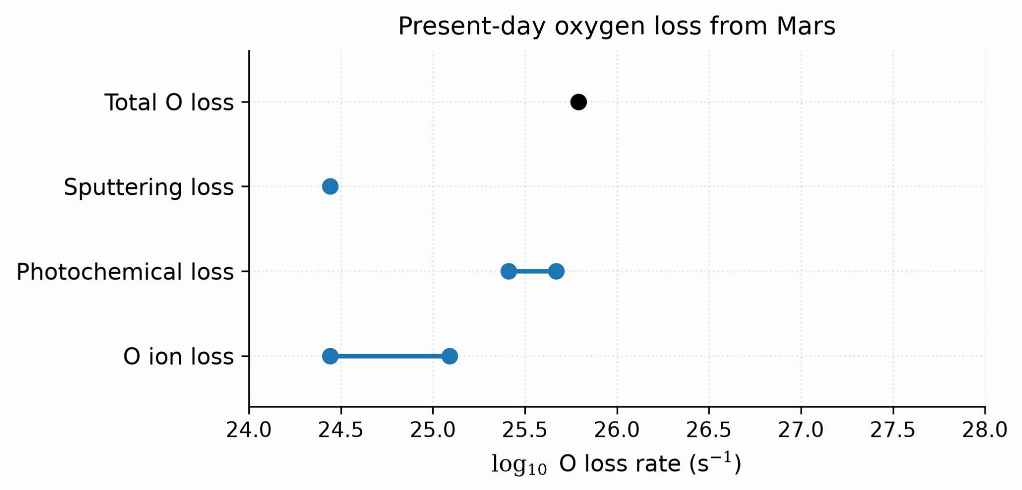
Sputtering: Solar wind ions strike upper atmosphere and eject neutral molecules

Photochemical: Photodissociation yields suprathermal atoms exceeding v_{esc}

Ion pickup: Photoions are swept away by interplanetary magnetic fields

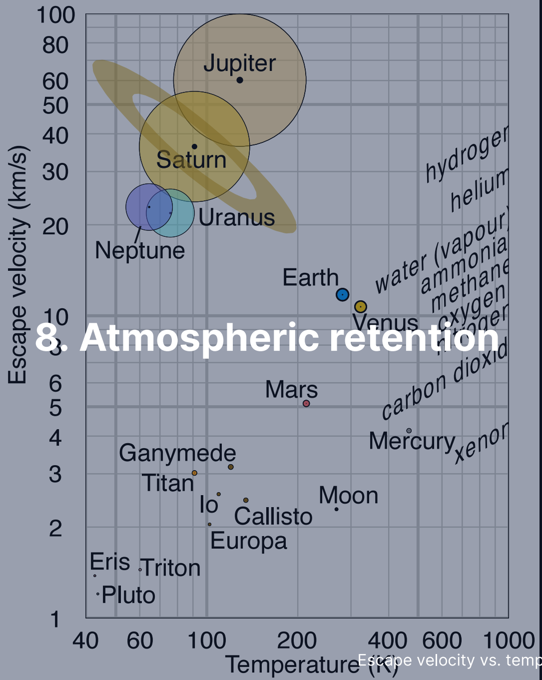
Impact erosion: Giant impacts shock and eject atmospheric columns into space

MAVEN at Mars: photochemical loss of oxygen dominates modern escape over sputtering and ion pickup.



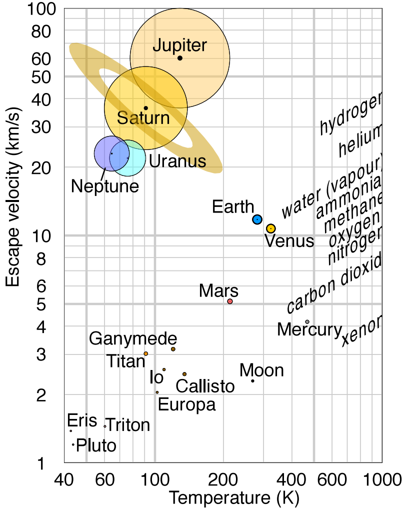
Present-day oxygen loss from Mars by escape channel: photochemical escape dominates modern oxygen loss; summed with hydrogen escape, the total reaches the 2–3 kg s^{-1} MAVEN measures. After Jakosky et al. (2018).

8. Atmospheric retention



Escape velocity vs. temperature. Credit: Wikimedia, CC BY-SA 4.0

Atmospheric retention: v_{esc} vs. T diagram



Credit: Wikimedia, CC BY-SA 4.0

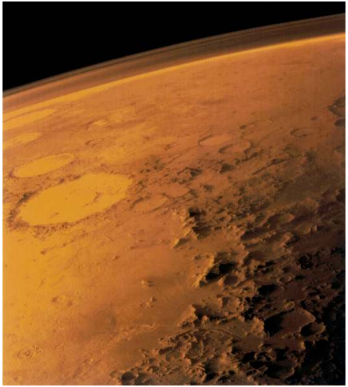
Rule of thumb for long-term retention:

$$v_{\mathrm{esc}} \gtrsim 6\, v_{\mathrm{th}}$$

- Gas giants: retain all species
- Earth/Venus: retain heavy species, lose H
- Mars: marginal; non-thermal losses dominate
- Titan: cold → retains N_2 despite low v_{esc}
- Moon/Mercury: retain no significant atmosphere

Mars: a lost atmosphere

Loss of Mars' magnetic dynamo allowed solar wind stripping to erode an early thick atmosphere down to 6 mbar.

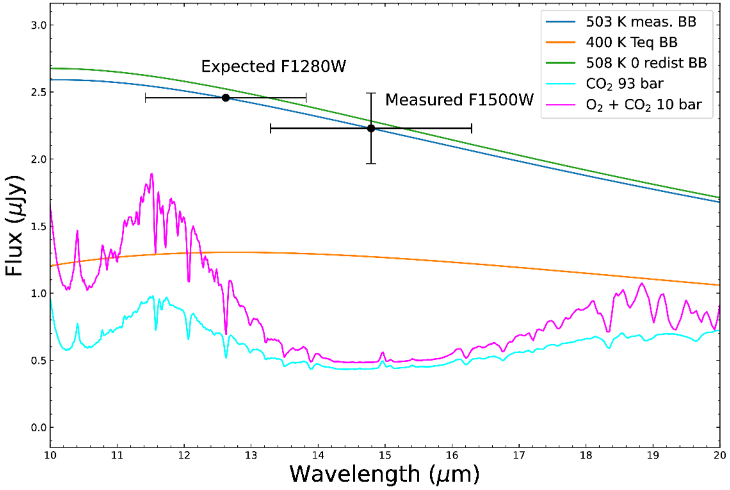


Credit: NASA/JPL (Viking 1, 1976)

- Today 6 mbar: no stable liquid water, little greenhouse warming
- Valley networks and lakes record a dense early atmosphere
- The dynamo died near ~4.1 Ga and left the atmosphere to the solar wind

9. Recent advances & outlook

Earth’s atmosphere. Credit: NASA/ISS



JWST/MIRI 15 μm thermal emission matches a 503 K bare rock, ruling out a thick atmosphere on TRAPPIST-1 b. Greene et al. (2023).

Summary

1. **Atmosphere types**: primary (captured), secondary (outgassed), tertiary (biogenic)
2. **Hydrostatic balance**: $\frac{\mathrm{d}P}{\mathrm{d}z} = -\rho g$ yields exponential pressure decay $P_0 e^{-z/H}$
3. **Scale height**: $H = k_\mathrm{B} T/(\mu m_u g)$ balances thermal energy against gravity
4. **Vertical structure**: convection sets the lapse rate ($\Gamma_d = g/c_p$ dry, less when condensing) below the 0.1 bar tropopause
5. **Beer-Lambert law**: $I = I_0 e^{-\tau}$; thermal radiation escapes from photosphere at $\tau \approx 1$
6. **Greenhouse warming**: optical depth raises surface temperature T_s above T_eff
7. **Atmospheric escape**: thermal, hydrodynamic, and non-thermal; retention requires $v_\mathrm{esc} \gtrsim 6 v_\mathrm{th}$

Next: Lecture 6

Questions?

Next: Lecture 6, Atmospheres II: clouds, weather & climate (Mara Attia)

Thu 24 Sep, 11:00–13:00, room 5419.0161 (Collegezaal)

Next tutorial: Fri 25 Sep, 13:00–15:00, room 5419.0161 (Collegezaal)

`https://ips.formingworlds.space`